1-14-2025

- Powered on LAZAM Ctrl-C
- Opened Terminal
- Entered ‘sudo apt install qv412’
- Entered ‘qv412’, terminal returned “command not found”
- Entered ‘sudo apt install qv4l2’
- Entered ‘qv4l2’, terminal opened V4L2 Test Bench
- Opened ‘video 0’ in device selection
- Hit ‘Start Capturing’
    - Input set to ‘Camera 1’
    - Frame Size set to 640x480
    - Frame Rate set to 30 fps
    - Capture Image Formats set to ‘YUYV 4:2:2’
    - Colorspace set to ‘Autodetect’
    - Y’CbCr/HSV Encoding set to ‘Autodetect’
    - Video Aspect Ratio set to ‘Source Width and Height’
    - Streaming Method set to ‘Memory mapped I/O’
    - Field set to ‘None’
    - Transfer Function set to ‘Autodetect’
    - Quantization set to ‘Autodetect’
    - Pixel Aspect Ratio set to ‘Autodetect’
    - Number of Buffers set to 4
    - Use Record Priority unchecked
- ArduCam successfully streamed video with minimal latency
- Closed V4L2 Test Bench
- Powered off LAZAM Ctrl-C

1-16-2025

- Powered on LAZAM Ctrl-C
- Opened Terminal
- Entered ‘git clone https://github.com/Tinker-Twins/AprilTag.git’
- Entered ‘cd ~/AprilTag’
- Entered ‘./install.sh’, failed to install due to lack of file permissions
- Entered ‘chmod +x install.sh’
- Entered ‘./install.sh’, installed successfully
- Entered ‘cd ~/AprilTag/scripts’
- Entered ‘python3 apriltag_video.py’, script successfully captured AprilTag data from preloaded video
- Closed Terminal
- Changed line 10 in python3 apriltag_video.py
    - Was: def apriltag_video(input_streams=['../media/input/single_tag.mp4', '../media/input/multiple_tags.mp4'],
    - Now: def apriltag_video(input_streams=[0],
- Opened Terminal

- Entered ‘cd ~/AprilTag/scripts’
- Entered ‘python3 apriltag_video.py’, script successfully captured AprilTag data from live ArduCam stream
    - Used image of multiple 11h36 AprilTags on a computer monitor
    - Printed 11h36 AprilTags 0-6 on white paper
- Closed Terminal
- Powered off LAZAM Ctrl-C